

Level 1 — Java Exercises

Problem 1 — Divisible by 5

Description Write a program to check if a number is divisible by 5 .

I/P => number

O/P => Is the number ___ divisible by 5 ? ___

```
import java.util.Scanner;

public class DivisibleByFive {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int number = sc.nextInt();
        System.out.println("Is the number " + number + " divisible by 5? " +
(number % 5 == 0));
    }
}
```

Problem 2 — First is the Smallest

Description Write a program to check if the first is the smallest of the 3 numbers.

I/P => number 1 , number 2 , number 3

O/P => Is the first number the smallest? ___

```
import java.util.Scanner;

public class FirstIsSmallest {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int number1 = sc.nextInt();
        int number2 = sc.nextInt();
        int number3 = sc.nextInt();
        boolean isSmallest = number1 < number2 && number1 < number3;
        System.out.println("Is the first number the smallest? " +
isSmallest);
    }
}
```

Problem 3 — Which One Is Largest

Description Write a program to check if the first, second, or third number is the largest of the three.

I/P => number 1 , number 2 , number 3

O/P =>

Is the first number the largest? ____
Is the second number the largest? ____
Is the third number the largest? ____

```
import java.util.Scanner;

public class WhichIsLargest {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int number1 = sc.nextInt();
        int number2 = sc.nextInt();
        int number3 = sc.nextInt();
        System.out.println("Is the first number the largest? " + (number1 >
number2 && number1 > number3));
        System.out.println("Is the second number the largest? " + (number2 >
number1 && number2 > number3));
        System.out.println("Is the third number the largest? " + (number3 >
number1 && number3 > number2));
    }
}
```

Problem 4 — Natural Number & Sum of First n Natural Numbers

Description Check for a natural number and print the sum of first n natural numbers if valid.

Hint A Natural Number is a positive integer (1 , 2 , 3 ,...). Use formula $n*(n+1)/2$ for sum.

I/P => number

O/P => If the number is a positive integer then the output is

The sum of ____ natural numbers is ____

Otherwise

The number ____ is not a natural number

```
import java.util.Scanner;

public class NaturalNumberSum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int number = sc.nextInt();
        if (number >= 1) {
            long sum = (long) number * (number + 1) / 2;
            System.out.println("The sum of " + number + " natural numbers is
" + sum);
        } else {
            System.out.println("The number " + number + " is not a natural
number");
        }
    }
}
```